A conditional split-module lift for Read's ordinary Calkin algebra

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Source question

The source paper is Richard Skillicorn, *The uniqueness-of-norm problem for Calkin algebras*, arXiv:1507.08118. Remark 2.6 asks whether the ordinary Calkin algebra

$$\mathcal{B}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}})$$

of Read's Banach space has a unique complete algebra norm.

Summary

This note records a conditional reduction, not a full answer. It proves an abstract criterion for lifting a non-equivalent complete norm from a square-zero split quotient through a Banach-algebra extension. In Read's case, Laustsen–Skillicorn prove a strongly split exact sequence

$$0 \longrightarrow \mathcal{W}(E_{\mathrm{R}}) \longrightarrow \mathcal{B}(E_{\mathrm{R}}) \longrightarrow \ell_2^{\sim} \longrightarrow 0,$$

which descends to a split quotient

$$0 \longrightarrow \mathcal{W}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}}) \longrightarrow \mathcal{B}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}}) \longrightarrow \ell_2^{\sim} \longrightarrow 0.$$

The remaining point is exactly the module obstruction: an exotic complete norm on the ℓ_2-coordinate must still make the induced left and right actions on $\mathcal{W}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}})$ bounded.

Proof Intuition

For the weak Calkin algebra, Skillicorn can renorm the square-zero ℓ_2-radical directly: a discontinuous linear bijection of ℓ_2 extends to an algebra automorphism of $\ell_2^{\sim}$, giving a new complete algebra norm. For the ordinary Calkin algebra one has an extra ideal $J = \mathcal{W}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}})$. In a split form the algebra is a semidirect product $J \rtimes \ell_2^{\sim}$. The same exotic norm lifts precisely when the ℓ_2-coordinate still acts boundedly on J. Thus the abstract criterion below separates the already-solved quotient renorming from the one missing Read-space calculation.

Abstract lifting criterion

Let E be a Banach space regarded as a Banach algebra with zero product, and let $\widetilde{E} = E \oplus \mathbb{K}1$ be its unitization. Suppose that

$$0 \longrightarrow J \longrightarrow A \xrightarrow{\pi} \widetilde{E} \longrightarrow 0$$

is a strongly split extension of Banach algebras, with continuous homomorphic right inverse $\rho : \widetilde{E} \to A$. We identify A algebraically with

$$J \oplus \rho(\widetilde{E}).$$

For $\xi \in E$ define left and right module maps on J by

$$L_\xi(j) = \rho(\xi)j, \qquad R_\xi(j) = j\rho(\xi).$$

Theorem 1 (Split-module norm lift)**.** *Assume that E admits a complete norm α which is not dominated by the original norm on E, and that there is a constant $C \geq 0$ such that*

$$\|L_\xi j\|_J + \|R_\xi j\|_J \leq C\,\alpha(\xi)\ \|j\|_J$$

for every $\xi \in E$ and $j \in J$. Then A admits a complete algebra norm not equivalent to its original Banach-algebra norm.

Proof. Put $\beta(\xi + \lambda 1) = \alpha(\xi) + |\lambda|$. Since E has zero product, β is a complete algebra norm on $\widetilde{E}$. Let M be a bound for multiplication in J, and choose constants $D \geq \max\{1, C\}$ and $L \geq M$. Define

$$\|j + \rho(u)\|_* = L\,\|j\|_J + D\beta(u) \qquad (j \in J,\ u \in \widetilde{E}).$$

As a vector-space direct sum of the Banach spaces J and $(\widetilde{E}, \beta)$, $(A, \|\cdot\|_*)$ is complete.

Write $u = \xi + \lambda 1$ and $v = \eta + \mu 1$. The product in the split decomposition is

$$(j + \rho(u))(k + \rho(v)) = jk + j\rho(v) + \rho(u)k + \rho(uv).$$

The J-part is bounded by

$$M\,\|j\|\,\|k\| + C\alpha(\eta)\,\|j\| + |\mu|\,\|j\| + C\alpha(\xi)\,\|k\| + |\lambda|\,\|k\|\,.$$

After multiplying by L, this is dominated by the J-cross terms in $(L\,\|j\| + D\beta(u))(L\,\|k\| + D\beta(v))$, using $L \geq M$ and $D \geq C, 1$. The quotient part satisfies

$$\beta(uv) \leq \beta(u)\beta(v)$$

because E has zero product. Since $D \geq 1$, the quotient contribution $D\beta(uv)$ is dominated by $D^2\beta(u)\beta(v)$. Hence

$$\|(j + \rho(u))(k + \rho(v))\|_* \leq \|j + \rho(u)\|_*\ \|k + \rho(v)\|_*\,,$$

so $\|\cdot\|_*$ is a complete algebra norm.

It remains to see that it is not equivalent to the original norm. If it were bounded above by a constant multiple of the original norm on A, then on the split copy of E we would have

$$D\alpha(\xi) = \|\rho(\xi)\|_* \leq C_0\,\|\rho(\xi)\|_A \leq C_0\,\|\rho\|\ \|\xi\|_E\,,$$

contradicting that α is not dominated by the original norm. Thus the new complete algebra norm is non-equivalent to the original one. □

Corollary 1 (Annihilator-kernel subcase)**.** *In the setting of Theorem 1, suppose the common annihilator*

$$N = \{\xi \in E : L_\xi = 0 = R_\xi\}$$

is an infinite-dimensional complemented closed subspace of E. Then A has a non-equivalent complete algebra norm.

Proof. Write $E = N \oplus F$ as a topological direct sum. Choose a discontinuous linear bijection $\gamma : N \to N$, and set

$$\alpha(n+f) = \|\gamma n\|_E + \|f\|_E \, .$$

This is a complete norm on E, not dominated by the original norm. Since N annihilates J on both sides, the module maps depend only on the F-coordinate, and hence are bounded by a constant multiple of $\|f\|_E \le \alpha(n+f)$. Theorem 1 applies. □

Read's ordinary Calkin algebra

Laustsen–Skillicorn's split theorem provides a continuous homomorphism

$$\psi : \mathcal{B}(E_{\mathrm{R}}) \to \ell_2^{\sim}$$

with kernel $\mathcal{W}(E_{\mathrm{R}})$ and a continuous homomorphic right inverse. Since $\mathcal{K}(E_{\mathrm{R}}) \subseteq \mathcal{W}(E_{\mathrm{R}})$, this descends to a strongly split extension

$$0 \to J \to A \to \ell_2^{\sim} \to 0, \qquad A = \mathcal{B}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}}), \quad J = \mathcal{W}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}}).$$

Consequently Theorem 1 gives:

Corollary 2 (Conditional negative answer to Skillicorn's Remark 2.6)**.** *Let $\rho : \ell_2^{\sim} \to \mathcal{B}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}})$ be the splitting induced by Laustsen–Skillicorn. If there is a complete norm α on ℓ_2, not dominated by the Hilbert norm, such that*

$$\|\rho(\xi)j\| + \|j\rho(\xi)\| \le C\alpha(\xi)\,\|j\| \quad (\xi \in \ell_2,\ j \in \mathcal{W}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}})),$$

then $\mathcal{B}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}})$ does not have a unique complete norm.

In particular, the same conclusion follows if the common annihilator in ℓ_2 of the two module representations on $\mathcal{W}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}})$ is infinite-dimensional.

Proof. Apply Theorem 1 to the displayed split extension. The final sentence follows from Corollary 1, since every closed subspace of the Hilbert space ℓ_2 is complemented. □

Conditional Dependencies

Theorem 1 and Corollary 1 are unconditional. The dependency for the source problem is the hypothesis in Corollary 2. The literature checked in this push gives the strong splitting and the square-zero quotient, but not a product statement such as

$$\rho(\ell_2)\,(\mathcal{W}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}})) = (0) \quad \text{or} \quad (\mathcal{W}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}}))\,\rho(\ell_2) = (0),$$

nor the weaker boundedness of those actions with respect to an exotic complete norm on ℓ_2. This is the obstruction left for verification.

Literature status

Skillicorn proves that the weak Calkin algebra $\mathcal{B}(E_{\mathrm{R}})/\mathcal{W}(E_{\mathrm{R}})$ has at least two non-equivalent complete algebra norms and states the ordinary Calkin question as open in Remark 2.6. Ware's thesis shows that the ordinary Read Calkin algebra does not have a unique algebra norm: the quotient norm is not maximal, via Read's discontinuous derivation. This does not produce a non-equivalent complete norm. Arnott–Laustsen later record that the weak Calkin algebra is neither maximal nor incompressible, again at the weak-Calkin level. No later source located in the present search settles the complete-norm question for $\mathcal{B}(E_{\mathrm{R}})/\mathcal{K}(E_{\mathrm{R}})$.

Verification notes

The previous naive norm $\|j\| + \beta(u)$ on $J \oplus \rho(\ell_2^\sim)$ fails without the module-boundedness hypothesis because the cross terms $\rho(\xi)j$ and $j\rho(\xi)$ may be unbounded in the exotic quotient coordinate. Theorem 1 isolates exactly the missing boundedness. Future verification should compute, using the explicit Laustsen–Skillicorn operators T_ξ, whether a large subspace of the ℓ_2-coordinate annihilates $J = \mathcal{W}(E_\mathrm{R})/\mathcal{K}(E_\mathrm{R})$ modulo compact operators, or whether the module representation factors through a weaker Banach norm on ℓ_2.

References

- R. Skillicorn, *The uniqueness-of-norm problem for Calkin algebras*, Math. Proc. R. Ir. Acad. 115A (2015), 145–152; arXiv:1507.08118.
- N. J. Laustsen and R. Skillicorn, *Extensions and the weak Calkin algebra of Read's Banach space admitting discontinuous derivations*, Studia Math. 236 (2017), 51–62; arXiv:1602.08963.
- G. K. Ware, *Uniqueness of norm properties of Calkin algebras*, PhD thesis, Australian National University, 2014.
- M. Arnott and N. J. Laustsen, *Uniqueness of algebra norm on quotients of the algebra of bounded operators on a Banach space*, arXiv:2308.11586.